Head-direction invariance, not information, determines whether a predictive cognitive map folds under symmetry

Manas Venkata Sai Ravulapalli[1]

[1]Ashoka University, Sonipat, India

***For correspondence:** manasvenkatasai.ravulapalli_ug2023@ashoka.edu.in (MR)

Abstract Predictive learning produces place-cell-like spatial codes in recurrent networks (***Levenstein et al., 2024***), but the arenas used to validate this are built so that every position looks unique. Real arenas are symmetric, and symmetric landmarks make distinct positions observationally identical. We train predictive recurrent networks in arenas with C_1, C_2 and C_4 rotational landmark symmetry and manipulate their head-direction input, comparing conditions *within* a single arena so the observation stream is held exactly fixed. The decisive contrast holds information constant and varies only symmetry: two head-direction encodings of exactly one bit each, one invariant under a $180°$ rotation and one not. In the C_2 arena the two separate completely in orbit-phase decoding ($p = 1.1 \times 10^{-5}$, $n = 10$ networks per encoding); in the asymmetric arena they are indistinguishable ($p = 0.97$). Invariance, not information, folds the code, and the C_4 arena approaches the group-theoretic $1/|G|$ ceiling. The effect requires prediction: a matched autoencoder folds every encoding ($p = 0.59$), and one step of prediction restores the dissociation. The same mechanism reproduces the place-field repetition seen across identical compartments in rodents (***Grieves et al., 2016***), and survives a head-direction signal corrupted on a third of its steps. A self-motion signal resolves only those symmetries under which it is not itself invariant, and only when the objective forces the network to integrate it.

Introduction

Levenstein et al. (***2024***) showed that a recurrent network trained to predict its own future observations develops spatially tuned units without any spatial supervision, and validated this in an L-shaped arena chosen so that every position has a unique visual fingerprint. Most rodent electrophysiology is done in square arenas, which are rotationally symmetric; the condition the validation rests on — global observational discriminability — does not hold there. Symmetric landmarks make distinct positions observationally identical, and the theoretical link between predictive learning and spatial coding, the successor representation (***Stachenfeld et al., 2017***), inherits the problem: positions related by a symmetry of the transition structure have identical expected future occupancy.

Something must break the tie. The obvious candidate is self-motion. The head-direction system tracks global orientation by path integration and is anchored to landmarks by vision (***Taube et al., 1990***; ***Knierim et al., 1995***); when landmarks are symmetric the two signals conflict. The standard reading is that head direction supplies *extra information*, and that more information means less ambiguity.

Table 1. The four head-direction encodings. *axis* and *parity* are matched at one bit and differ only in invariance.

encoding	bits	C_2-invariant	C_4-invariant
full (E,S,W,N)	2.0	no	no
parity ({E,S} vs {W,N})	1.0	no	no
axis ({E,W} vs {N,S})	1.0	**yes**	no
const	0.0	yes	**yes**

We show that this reading is wrong, and that the correct variable is not the quantity of information but its *symmetry*. If the head-direction encoding ϕ satisfies $\phi(h+g) = \phi(h)$ for the generator g of the arena's symmetry group G, then the entire (observation, action) input process is G-equivariant, the posterior over position is G-invariant, and no amount of that signal can lift the ambiguity — the code must collapse onto the quotient X/G. An encoding carrying exactly the same number of bits, but not invariant under g, lifts it completely.

The same design yields a sequence of further results. The fold is visible directly in single-cell tuning: the place fields the networks grow without spatial supervision repeat at symmetry-related locations when the compass is invariant, once under C_2 and four times under C_4. Read through the C_4 quotient, the folding produces a hard $1/|G|$ ceiling that the networks approach to within a few percent. The effect is a property of *predictive* learning, since a matched reconstruction objective folds every encoding alike, and it survives a compass corrupted on a third of its steps. It also reproduces an experiment: place cells in visually identical compartments repeat when the compartments are translationally related and separate when they are rotationally related, the dissociation ***Grieves et al.*** (***2016***) reported in the rodent hippocampus. Throughout we distinguish a folded code from a broken one; the folded networks build a reproducible spatial map and merely discard which copy of the fundamental domain the animal occupies.

Results

A design that holds information constant and varies only symmetry

Networks receive a $7{\times}7{\times}3$ egocentric view and a five-dimensional SpeedHD action vector, $[\,\text{speed}, \text{onehot}(h)$ with $h \in \{\text{E}, \text{S}, \text{W}, \text{N}\}$. We replace the heading block by $M\,\text{onehot}(h)$ for a 4×4 matrix M, preserving dimensionality and column sums (Table 1). Two of the four encodings partition the headings into two classes of two — exactly one bit each, identical entropy, identical activation statistics — and differ only in *which* partition, and hence in whether they are invariant under the $180°$ rotation R^2.

The two hypotheses make opposite predictions. If folding tracks how much information the compass carries, *axis* and *parity* behave alike (Fig. 1a). If it tracks invariance, *axis* folds and *parity* does not.

Predictive learning grows place cells, and symmetry makes them repeat

Trained on prediction alone, the networks develop units with localised spatial firing fields, the place cells reported for this objective by ***Levenstein et al.*** (***2024***) (Fig. 1b). In the asymmetric arena under the full compass, about 480 of the 500 units carry a place field (a median of about two fields per unit). The effect of symmetry is visible in the rate maps before any decoder is applied. When the arena is symmetric and the head-direction encoding is invariant to that symmetry, the fields repeat: a unit in the C_2 arena under *axis* fires at a location and again at its $180°$ image, and a unit in the C_4 arena under *const* fires at all four $90°$ copies. A single unit reports every member of a symmetry orbit as the same place. The sections that follow quantify this and isolate what controls it.

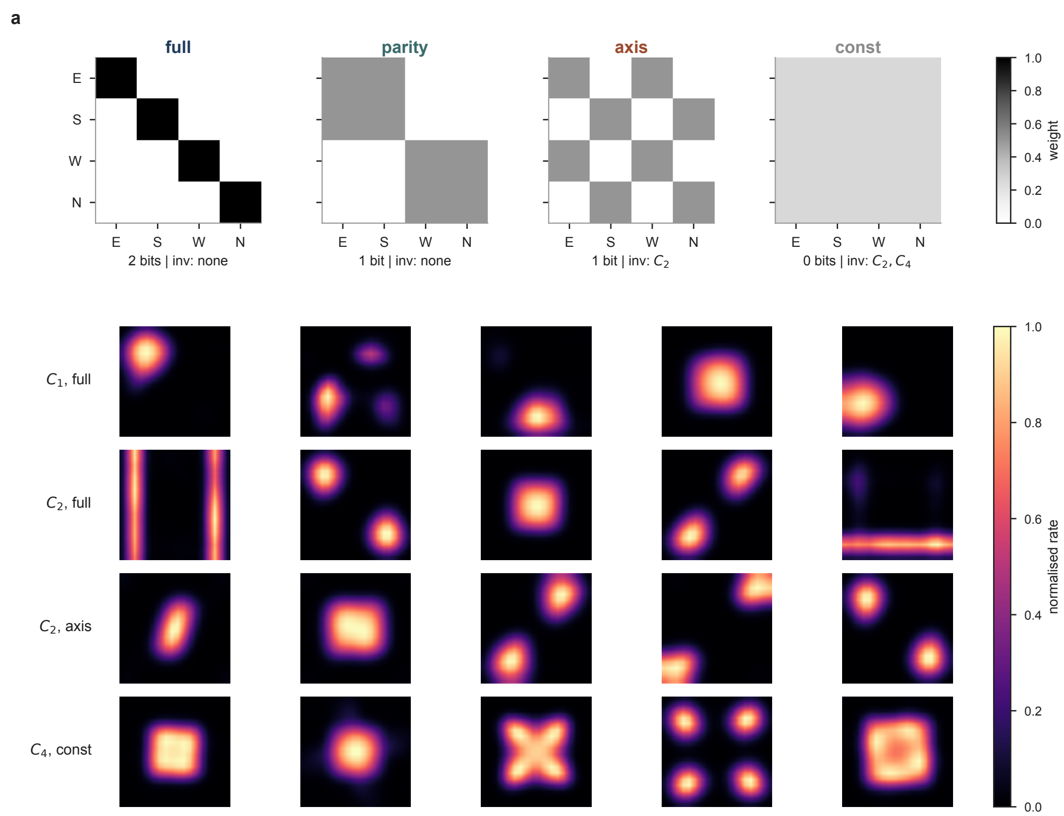

Figure 1. Design and phenotype. (**a**) The four head-direction encodings, as 4×4 transforms of the one-hot heading block {E,S,W,N}; grey level is the matrix weight. *full* is the identity (two bits); *parity* and *axis* collapse the compass to one bit with different partitions; *const* discards it. *axis* and *const* are invariant under the $180°$ rotation and fold the C_2 arena; *const* is the only encoding invariant under the full C_4 group. *axis* and *parity* carry identical information and differ only in invariance. (**b**) Mean firing-rate maps (smoothed, peak-normalised) of example units. Fields are single and sharp when the code does not fold, and repeat at the symmetry-related locations — once under C_2 (*axis* encoding) and four times under C_4 (*const*). Units are representative; the complete unselected set for each condition is given in the Supplementary Information.

Invariance, not information, folds the code

Under folding onto X/G the population state satisfies $h(x) = h(R^2x)$, so no decoder can recover which element of the orbit $\{x, R^2x\}$ produced it. We therefore train a linear decoder for *orbit phase*, holding out whole orbits across cross-validation folds so the decoder must generalise a phase direction to positions it has never seen. Chance is 0.500.

In the C_2 arena, *axis* and *parity* — carrying identical information — differ by 0.403 in phase accuracy. The separation is complete: every *parity* network scores above every *axis* network (Mann–Whitney $U = 0$, $p = 1.1 \times 10^{-5}$, the exact two-sided floor at $n = 10$ vs. 10; $p = 3.2 \times 10^{-5}$ after Bonferroni correction across the three arenas). In the C_1 arena, where there is no symmetry to collapse onto, the same two encodings are indistinguishable ($\Delta = 0.000$, $p = 0.97$). The C_1 null is what licenses the causal reading: *axis* does not degrade spatial coding in general, it degrades it exactly when the arena has a symmetry under which *axis* is invariant.

The chance-level reading of the folded codes is not an artefact of using a linear decoder. On the same population states, a k-nearest-neighbour decoder and a multi-layer perceptron also sit at chance for the invariant encodings in the C_2 arena (*axis*: 0.537 and 0.535; *const*: 0.531 and 0.531; $n = 6$), and both recover orbit phase for the non-invariant encodings (*full* and *parity*, every network ≥ 0.90). The orbit-phase information is absent from the folded code, not merely inaccessible to a linear readout.

None of the folded networks has simply stopped representing space. For 104 of the 112 net-

Table 2. Orbit-phase decoding accuracy (chance $= 0.500$). $n = 10$ per cell for s1 and s2, $n = 8$ for s4.

arena	*full*	*parity*	*axis*	*const*
s1 (C_1, no symmetry)	0.981	0.972	0.971	0.966
s2 (C_2)	0.978	0.955	**0.552**	**0.552**
s4 (C_4)	0.984	0.933	**0.544**	**0.521**

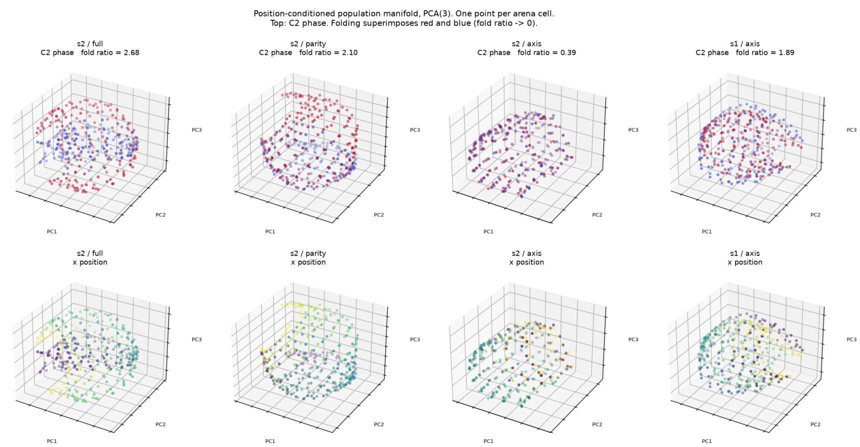

Figure 2. The fold, seen geometrically. Position-conditioned population manifold of a representative network (one point per arena cell, PCA to three dimensions), coloured by C_2 orbit phase. Under *full* and *parity* the two phases occupy separate regions; under *axis* in the C_2 arena they superimpose. The fold ratio — distance between a cell and its $180°$ image, in units of the distance to a spatial neighbour — is 2.68, 2.10 and 0.39 for the three C_2-arena encodings shown, and 1.89 for *axis* in the C_1 arena, where the same encoding folds nothing. The ratio is a property of the full hidden space, not of the linear projection: the folded *axis*/C_2 case reads below one and every non-folded case above one under distance-preserving nonlinear embeddings (Isomap, t-SNE) as well as PCA. We reduce with PCA because the claim is metric — two positions coincide — and unlike neighbour-graph embeddings PCA preserves distances rather than reshaping them.

works, $\max(R^2_{\text{raw}}, R^2_{\text{domain}}) \geq 0.81$: the folded codes still linearly encode position *within the fundamental domain*, they have merely discarded which copy of it the animal occupies (Fig. 2). The eight exceptions are *const* in the C_4 arena, which fails both C_2-based readouts ($R^2_{\text{raw}} = -0.069$, $R^2_{\text{domain}} = -0.589$) not because it has stopped coding space but because it is folded by the *full* C_4 group, which a C_2 decoder cannot see; the next section resolves it.

The C_4 arena reproduces the group-theoretic ceiling

A code folded onto X/G cannot name the element of an orbit, only the coset, so a $|G|$-way phase decoder is bounded above by $1/|G|$. Reading the C_4 arena through its own quotient (four-way phase, chance 0.250) recovers this pattern (Table 3). *const* is the only C_4-invariant encoding, and the only one to fall to the C_4 floor (0.275, just above 0.250). *axis* is C_2- but not C_4-invariant, and sits near the C_2 ceiling of 0.5 (0.482). The folded values sit within a few percent of their predicted floors rather than exactly on them; the small residual is consistent with finite-sample decoding, and the decisive quantity is which floor each encoding falls to, not the residual.

This also resolves *const* in the C_4 arena. Read through the C_4 quotient it is plainly a live spatial code — its C_4 fundamental-domain R^2 is $+0.646$ (minimum $+0.535$ across all sixteen s4 networks) — and its four-way phase accuracy of 0.275 sits at the C_4 chance floor. What looked like a dead code under a C_2 readout is a code folded onto a larger quotient than the readout was built to see. We flag this because a G-folded code and a code that has stopped representing space are easy to

Table 3. C_4 four-way phase decoding in the s4 arena (chance = 0.250).

encoding	invariance	measured	predicted ceiling
full	none	0.918	—
parity	none	0.860	—
axis	C_2 only	**0.482**	0.500
const	C_2 and C_4	**0.275**	0.250

Table 4. The bound of Eq. 1, computed from the arenas with no free parameters, against the trained networks. Every encoding falls to the accuracy its input-distinguishability predicts; the mean absolute error across the twelve cells is 0.031. (We report the bound as a bound, not a fit: the predicted values are near-degenerate — clustered at 1.0 and at the folding floors — so a correlation coefficient across cells would merely register that separation.)

arena	encoding	distinguishable	predicted	measured
s1	*full* / *parity*	1.000	1.000	0.981 / 0.972
s1	*axis* / *const*	0.934	0.967	0.971 / 0.966
s2	*full* / *parity*	1.000	1.000	0.978 / 0.955
s2	*axis* / *const*	0.000	0.500	0.552 / 0.552
s4	*full* / *parity*	1.000	1.000	0.984 / 0.933
s4	*axis* / *const*	0.000	0.500	0.544 / 0.521

confuse, and only the second is a failure.

The ceiling is set by the inputs; the objective decides whether it is reached

None of this requires a trained network to predict. Call two states (x, h) and (x', h') *input-equivalent* if they produce the same 7×7 observation and the same HD encoding. A code is a function of its inputs, so no decoder can separate input-equivalent states, and phase decoding is bounded above by

$$\text{acc}_{\text{max}} \;=\; \tfrac{1}{2} + \tfrac{1}{2} \Pr_{(x,h)}\big[\,(x,h) \nsim (R^2 x,\; h+2)\,\big], \tag{1}$$

the fraction of orbit pairs the input distinguishes. Evaluating the right-hand side on the arenas themselves — enumerating all 324×4 states and hashing their observations — takes no training at all.

Table 4 also settles a number that would otherwise look like a failure. In the C_1 arena *axis* reaches 0.971 rather than 1.000 — because 6.6% of states there are *accidentally* observationally identical to their C_2 partner, putting the bound at 0.967. *axis* and *const* sit exactly on it. They were not failing to fold; they were folding on precisely the ambiguous fraction.

The objective selects the bound.

The inequality is an upper bound, not a prediction: a network is free to discard information its loss does not reward. This is what the horizon sweep shows. At $k = 0$ the input still separates 1228 of the 1296 states under *full* — the compass is right there — yet all four encodings collapse to within 0.005 of one another (Table 5), landing on the *observation-only* bound of 0.500. The spread across encodings is 0.0052 at $k = 0$ against 0.427 at $k = 5$ (Fig. 3a).

So the compass is not absent under reconstruction; it is *discarded*. Reconstructing o_t never requires knowing which of two identical-looking places you occupy. Predicting o_{t+1} does, and one step is enough.

Table 5. Orbit-phase decoding in the C_2 arena as a function of prediction horizon k (chance = 0.500, $n = 6$ per cell).

k	*full*	*parity*	*axis*	*const*	*axis* vs *parity*
0 (autoencoder)	0.536	0.541	0.539	0.539	$p = 0.59$
1	0.975	0.941	0.545	0.546	$p = 0.00216$
3	0.968	0.947	0.548	0.545	$p = 0.00216$
5	0.978	0.953	0.553	0.551	$p = 0.00216$

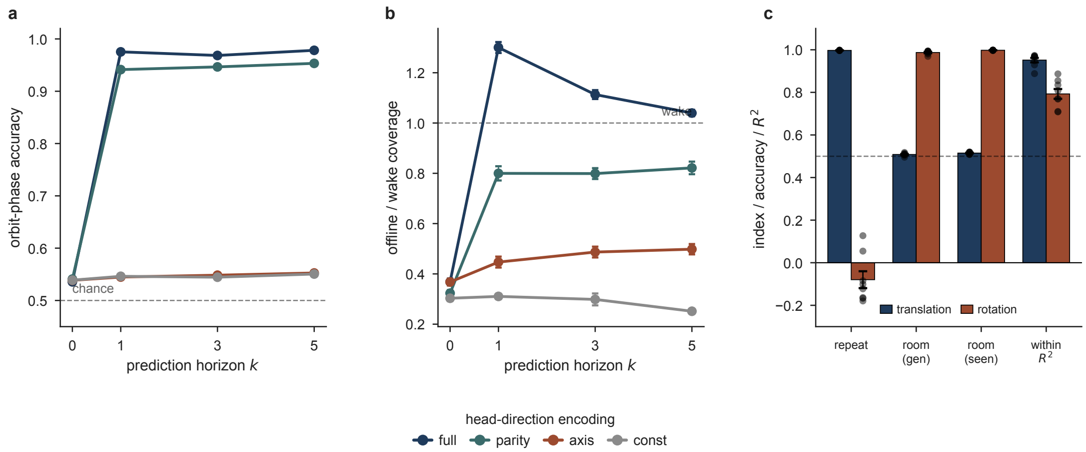

Figure 3. Functional consequences of the fold. (**a**) Orbit-phase decoding accuracy vs prediction horizon k in the C_2 arena (chance 0.5); the non-invariant encodings (*full*, *parity*) resolve the fold at $k \geq 1$, the invariant ones (*axis*, *const*) stay near chance. (**b**) Coverage of the self-generated offline trajectory as a fraction of a wake path of equal length (dashed line, wake). Both step up at $k = 1$ and are ordered by how much of the compass the encoding keeps; points are mean ± s.e.m., $n = 6$. (**c**) Place-field repetition across two observationally identical compartments ($n = 8$ each): under translation the fields repeat and the room is undecodable (chance 0.5), under rotation the fields do not repeat and the room decodes; within-room position (R^2) stays high in both. Dots are individual networks.

The fold is resolved by prediction, not by observation

We repeated the s2 sweep varying only the rollout horizon k. At $k = 0$ the network's target is its own current observation at the anchor (verified exactly), i.e. a pure autoencoder with the same architecture and the same head-direction input, but no future to predict.

Under reconstruction the full two-bit compass buys nothing: every encoding folds, and all four sit near chance (Table 5). These are folded codes, not broken ones — the minimum of $\max(R^2_{\text{raw}}, R^2_{\text{domain}})$ across all 24 autoencoder networks is 0.858. A single step of prediction restores the effect in full, and extending the horizon to five adds nothing: the dependence on k is a step, not a gradient.

The reason is that the autoencoder's target is the current observation, which the compass cannot help predict, so the objective gives the network no reason to integrate self-motion and the C_2-symmetric observation stream determines the code. Prediction of a future observation requires integrating self-motion across at least one step, and it is that integration which resolves the symmetry. The control therefore isolates a property of the objective rather than of reconstruction as such: an objective that does not require self-motion integration does not recruit the compass, whatever else it reconstructs.

Offline replay emerges at the same horizon as symmetry resolution

A predictive network can be run offline, driven by its own predicted observation in place of a real one, so that it generates trajectories without moving through the arena (***Levenstein et al., 2024***). We generated 200-step offline trajectories from each C_2 network and decoded position with a ridge decoder fit on waking activity (Fig. 3b). For each trajectory we measured the fraction of the arena it covered, relative to a waking trajectory of the same length; a 200-step walk visits only 27% of the arena, so the raw fraction is hard to read on its own.

Offline coverage rose sharply between $k = 0$ and $k = 1$ and changed little afterwards, tracking the step in orbit-phase accuracy over the same range. At $k = 0$, where the network reconstructs its current observation, coverage stayed below 37% of waking under every encoding. From $k = 1$ the *full* compass drove trajectories that covered more of the arena than waking behaviour itself, at 1.30, 1.11 and 1.04 times waking for $k = 1$, 3 and 5. Coverage fell as the compass was weakened, from 1.30 times waking under *full* to 0.80, 0.45 and 0.31 under *parity*, *axis* and *const*, the same order in which the code folds. Step-to-step continuity did not follow this ordering and was highest for *const*, since a network whose state barely moves produces short smooth steps whatever it represents. Offline generation and symmetry resolution therefore depend on prediction and on head direction in the same way, and appear at the same horizon.

These offline trajectories are temporally ordered rather than random walks, and only when the compass is intact. We score sequenceness as the rank correlation between a rollout's principal-axis position and time, against a circular time-shift null and a cell-identity shuffle. In the C_2 arena the *full* network's offline paths reach a sequenceness of 0.53, close to the waking reference of 0.56 and well above the cell-shuffle floor of 0.25, whereas the folded *axis* and *const* encodings fall to that floor (0.20 against 0.17). Coherent replay, like coverage, requires the head-direction signal the fold discards.

The dissociation survives a noisy compass

The head-direction signal used so far is a noiseless oracle, whereas the biological compass drifts and is re-anchored by vision. To test whether the fold depends on a perfect signal, we retrained the C_2 networks with the reported heading corrupted: on 30% of steps the compass was rotated by a random $\pm 90°$ before encoding. The dissociation is preserved. Orbit-phase decoding is 0.556 under *axis* and 0.922 under *parity*, a separation of 0.367 that remains complete: every *parity* network scores above every *axis* network (Mann–Whitney $U = 0$, $p = 0.0022$ at $n = 6$ vs. 6). Relative to the clean networks the separation narrows only slightly (from 0.403), driven by a small drop in *parity* resolution ($0.955 \rightarrow 0.922$) while the folded *axis* code is unchanged ($0.552 \rightarrow 0.556$). A compass corrupted on a third of steps still folds the same symmetries it does when clean. (These networks were trained for 30,000 steps rather than 80,000; the effect is already complete by then.)

Two natural readouts fail, in two different ways

Decomposing each unit's rate map into the four characters of C_4 gives an isotypic power spectrum $P_0 \dots P_3$. Rotational autocorrelation, the quantity used in earlier analyses of this system, is exactly RA $= P_0 - P_2$. It is therefore *blind to* C_2 *structure*: invariance under R^2 annihilates the odd characters P_1 and P_3 while leaving P_0 and P_2 untouched, so a perfectly C_2-folded code and pure noise both read RA ≈ 0. Empirically, on a clean 17-network sweep, RA cannot separate s2 from s1 ($p = 0.056$), whereas odd power $P_1 + P_3$ separates them completely ($p = 0.0079$, the exact floor at $n = 5$ vs. 5).

Odd power, however, is itself confounded. Under *axis* it falls in the C_1 arena too, where there is no symmetry: the drop odd(*parity*) − odd(*axis*) is 0.095 in s1, 0.111 in s2 and 0.112 in s4, so 85% of the s4-sized effect is present where nothing can fold, and it separates completely in s1 ($p = 1.8 \times 10^{-4}$) — a confident report of a fold that does not exist. Odd power conflates genuine collapse onto the quotient with the encoding merely reshaping the rate-map spectrum. Orbit-phase decoding is immune to that and returns a clean null.

We therefore report phase decoding as the primary readout and the isotypic spectrum as a secondary, descriptive one. We note this explicitly because the confounded readout would have supported the same qualitative conclusion, by an argument that does not hold.

Folding leaves a reproducible spatial map intact

Because the folded encodings decode orbit phase near chance (about 0.55, reliably just above the 0.5 floor but far below the ~ 0.95 of the non-invariant encodings), we asked whether they had learned a spatial map at all, or whether ablating the compass had degraded the representation more broadly. Two measures argue that the map is intact (Fig. 4c,d). Spatial representational similarity, the Spearman correlation between neural distance and Euclidean distance across pairs of positions, fell as the compass was ablated, from 0.70 under *full* to 0.32 under *const* in the C_1 arena, and lower still in the symmetric arenas. This is the expected consequence of folding: a position and its group image sit on top of one another in the neural space while lying far apart in the arena, weakening the correlation between the two distances. The correlation between the position-conditioned codes of independently trained networks behaved oppositely and stayed above 0.96 under every encoding in every arena, and 448 of the 500 units retain a place field under *const* in the C_1 arena, falling to about 318 in the C_4 arena. A folded code is therefore reproducible rather than degenerate, and the compass governs how the map is laid out rather than whether one forms.

The collapse can also be read in the metric used to score remapping in the hippocampus. The population-vector correlation between a position and its symmetry image (***Leutgeb et al., 2005***) — near one when the two share a representation, near zero when the code separates them — is 0.98 under the invariant encodings in the C_2 and C_4 arenas and about 0.28 under *full*, with the matched non-invariant encoding intermediate (*parity*, 0.58 in C_2; $n = 10$ per encoding). By this measure the network fails to remap between symmetry-related locations exactly when the compass is invariant under the symmetry — the two copies are written as one place — whereas with a full compass they are separated as strongly as unrelated positions. The same invariant encoding gives a weaker correlation in the C_1 arena (0.71 for *axis*), where there is no arena symmetry to complete the fold.

Folding multiplies place fields and imprints the arena's symmetry on each map

The fold is visible cell by cell, not only in the population decoder. We characterised every unit's firing-rate map and counted its place fields (Fig. 4a). Fields per unit rise with both arena symmetry and encoding invariance, from 1.78 under *full* in the C_1 arena to 3.39 under *const* in the C_4 arena. On its own this count is a confounded readout: it rises even in the C_1 arena, where nothing can fold (2.44 fields under *const* against 1.78 under *full*), because a more invariant encoding yields more multi-peaked units regardless of symmetry. The fold-specific question is whether the extra fields sit at group-related positions. We measure this as the correlation between a unit's rate map and its own 180° rotation, a per-cell index of C_2 symmetry (Fig. 4b). The index saturates near 0.97 exactly when the encoding is C_2-invariant and the arena is symmetric (*axis* and *const* in the C_2 and C_4 arenas), while the matched non-invariant encoding stays low (*parity*, 0.36 in C_2) and *full* stays low throughout (≤ 0.16). An invariant encoding alone induces partial symmetry even in the C_1 arena (0.61 for *axis*), which the arena's own symmetry then completes. The four-fold version of the index reaches 0.97 only for *const* in the C_4 arena, the single fully C_4-invariant condition. Each map acquires precisely the symmetry the compass cannot break.

Scored by the criteria applied to real recordings, the trained population is a realistic mixture rather than a bank of identical units: 76–96% of units qualify as place cells (Skaggs spatial information above threshold) and about 30% are border-modulated (a third or more of their rate-map mass in the outer ring), a composition that is stable across encodings and arenas. Folding therefore multiplies fields and rearranges where they sit without changing the cell-type makeup of the map — the multi-field, spatially repeating cells it produces are the in-silico counterpart of the repeated place fields recorded in symmetric environments (***Muller and Kubie, 1987***; ***Grieves et al., 2016***).

The same dissociation appears across observationally identical compartments

Head direction should resolve rotational but not translational symmetry, and this can be tested in the geometry used to study repeated place fields in rodents. We built two closed 6×6 com-

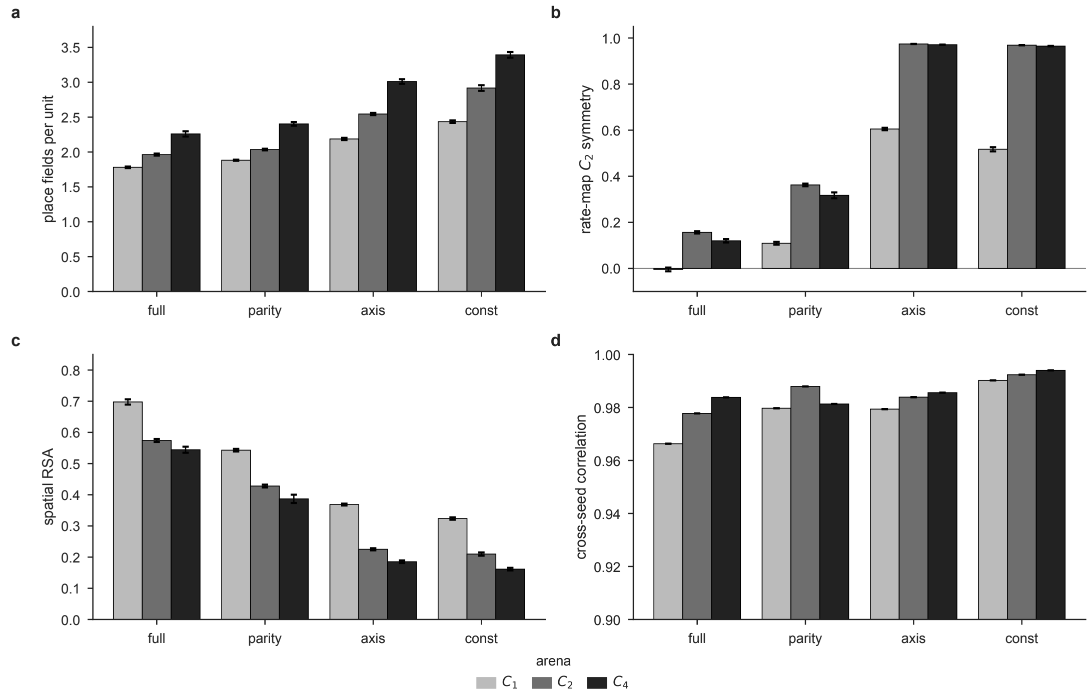

Figure 4. Quantifying the fold. (**a**) Mean place fields per unit; rises with both arena symmetry and encoding invariance, but also in the C_1 arena where nothing folds, so field count alone is confounded. (**b**) Rate-map C_2 symmetry index (correlation of a unit's map with its $180°$ rotation), the fold-specific readout; it saturates near 0.97 only when the encoding is invariant to the arena's symmetry. (**c**) Spatial representational similarity falls as the encoding is ablated and as arena symmetry increases, the signature of collapse onto the quotient. (**d**) Cross-seed correlation between independently trained networks stays above 0.96 throughout (y zoomed to 0.9–1.0): the fold is reproducible, not degenerate. Arenas $C_1/C_2/C_4$; bars are mean ± s.e.m. over networks ($n = 10$ per encoding for C_1 and C_2, $n = 8$ for C_4).

partments joined by a corridor and gave them identical interior landmarks, so that the two rooms produced the same set of egocentric views; across all 144 combinations of location and heading the observations coincided exactly. In one arrangement the rooms were related by a translation, under which the compass reads the same in both; in the other by a rotation, under which it does not. We trained networks on each arrangement and, for every network, decoded which room the animal occupied and measured whether individual units repeated their firing fields between the rooms (Fig. 3c).

The two arrangements gave opposite results ($n = 8$ networks each). Under translation the fields repeated almost exactly across the rooms (repetition 0.997) and the room could not be decoded even by a nearest-neighbour classifier tested on cells it had already seen (0.515, chance 0.5), so the network folded the two identical rooms onto one map. Under rotation the repetition vanished (repetition -0.080) and the room decoded perfectly (0.998). Position within a room decoded well in both arrangements ($R^2 = 0.95$ and 0.79), so neither network had stopped representing space. Every translation network separated from every rotation network on all four measures (Mann–Whitney $p = 1.6 \times 10^{-4}$, the exact two-sided floor at $n = 8$ vs. 8). ***Grieves et al.*** (***2016***) reported this same contrast in CA1, and here it follows from nothing but the group action of the arena on the compass.

Discussion

The result is a statement about group actions rather than about information channels. A self-motion signal resolves exactly those symmetries under which it is not itself invariant. Head direction is blind to translation — translating the world does not change a compass reading — and therefore cannot disambiguate translationally repeated compartments, whereas it is not blind to rotation and can disambiguate rotationally related ones.

The dissociation exists in the animal data

Spiers et al. (***2015***) recorded CA1 place cells in four visually identical compartments arranged in a row off a connecting corridor, and found “a high degree of spatial repetition with a slight degree of rate-based discrimination”. Those compartments are translationally related, and the authors conclude that “path integration is not used, or at least not spontaneously” to separate them. ***Grieves et al.*** (***2016***) then ran the same four compartments in two arrangements. Parallel, all facing the same way and therefore translationally related: “place cells often exhibited repeated fields”. Radial, at $60°$ to one another and therefore rotationally related: “significantly less place field repetition was apparent” (all $p < 0.0001$). The dissociation survived removal of the orientation landmark, so it is not carried by extra-maze vision, and the authors attribute it to “directional information derived from the animal’s self-motion”, hypothesising the head-direction system as the source.

Our result supplies the reason. A compass is *invariant* under translation and *equivariant* under rotation. The input process is therefore G-equivariant in the parallel arrangement and not in the radial one, so a predictive code must fold onto the quotient in the first case and may lift it in the second. Repetition is not a failure of the hippocampus; it is what a predictive code must do when its self-motion signal cannot see the symmetry.

Two things follow that the animal data cannot establish. First, the effect is a property of *invariance*, not of how much the compass tells you: two encodings carrying the same single bit dissociate completely (Table 2). Second, it is a property of *prediction*: under a matched reconstruction objective the compass is inert (Table 5), so head direction cannot disambiguate compartments unless the objective forces the network to integrate it. Whether a purely reconstructive circuit would show repetition even in the radial arrangement is, to our knowledge, untested in animals.

The account also yields a lesion prediction, with a dose attached. Lesions of the anterodorsal thalamus degrade the head-direction signal and destabilise hippocampal place fields (***Calton et al., 2003***). We predict that degrading the compass should bring field *repetition* into the *radial* arrangement, where intact animals show little, because without direction rotation joins translation as a symmetry the animal cannot resolve. The robustness control sets the threshold high, though: in the model a compass corrupted on 30% of steps still resolves the fold, and full repetition appears only in the *const* limit of near-total directional loss (which folds the C_2 arena to near chance, Table 2). A partial ADN lesion should therefore produce partial repetition, graded by how much of the head-direction signal survives.

What the model reproduces, and what it does not

The two rows marked *absent* are architectural, not empirical: head direction enters this network as an oracle input rather than being learned, and no non-negativity constraint is imposed, which is the condition under which hexagonal grids emerge in trained path integrators (***Sorscher et al., 2023***). Nothing here bears on the ring or toroidal manifolds of entorhinal cortex, and we make no claim about them.

The objective control constrains the mechanism further: a matched autoencoder with the same wiring and the same compass folds every encoding alike ($p = 0.59$), which rules out the fold being a property of the input pipeline rather than of the learning. It is an objective that requires predicting a future observation, not reconstructing the present one, that recruits the compass.

Relation to other models of the spatial code

The parallel-versus-radial dissociation is a discriminating test among models of place-cell firing, because the two arrangements are built from the same compartments and are locally identical in both. A model that fixes firing from local sensory or boundary geometry alone must therefore predict the same repetition in each: the boundary-vector account (***Hartley et al., 2000***), in which a field is a fixed function of the distances to walls, gives identical fields wherever the local geometry recurs, parallel or radial. A reconstruction objective behaves the same way for the same reason, and our autoencoder control confirms it — with the compass held identical it folds every encoding alike. Continuous-attractor path integrators (***Burak and Fiete, 2009***; ***Sorscher et al., 2023***) build the metric in by construction and do not speak to what happens when observations alias. The successor representation (***Stachenfeld et al., 2017***; ***Momennejad et al., 2017***) is predictive, but over a state space that is supplied rather than inferred from aliased observations, so whether two symmetric compartments are one state or two is an assumption, not a result (Table 7).

What predictive learning adds is a derivation. Trained only to predict its next view from self-motion, the network folds or lifts the identical compartments according to whether the compass is invariant under the arrangement's symmetry, reproducing the animal dissociation from a single group-theoretic principle rather than fitting it. The contribution is therefore not that prediction grows place cells (***Levenstein et al., 2024***), but that it specifies *when* the spatial code collapses under symmetry and when it does not — a question the other model classes leave open.

Limitations.

The arenas are discrete 18×18 grids with four headings, so the head-direction "ring" has four points and no continuous ring topology; nothing here speaks to the ring or toroidal manifolds reported in entorhinal cortex (***Gardner et al., 2022***). The networks develop place-like but not grid-like tuning (gridness over the 100 most spatially tuned units: mean -0.35, maximum $+0.08$, none above the conventional 0.3 threshold), which is expected given that head direction is supplied as an input rather than learned and that no non-negativity constraint is imposed (***Sorscher et al., 2023***); the toroidal code of ***Gardner et al.*** (***2022***) cannot exist in this model, and we do not claim it. The head-direction signal is a noiseless oracle, whereas the biological compass drifts and is re-anchored by vision; a control that corrupts the compass on 30% of steps (Results) leaves the dissociation intact, but a fully learned, continuously drifting compass is not tested here. We also tested whether population-manifold topology crystallizes before metric geometry, a proposed developmental ordering. With a shuffle-null Betti estimator the effect does not hold in these arenas: position is linearly decodable at $R^2 = 0.79$ in the untrained network, metric geometry develops gradually. Loop topology is readout-dependent: the linear PCA-6 estimator does not reliably recover it, yet a spectral (Laplacian) embedding recovers the annulus's single loop in every network and the open arena's trivial topology, both under-detected by PCA-6 (`topology_robustness.csv`). The two-loop theta and figure-8 arenas are not cleanly recovered by any embedding — the higher-dimensional linear reduction that occasionally finds them also hallucinates a loop in the open arena. Simple loop topology is thus present but visible only through a nonlinear readout, and higher-genus topology is not reliably recovered, so we make no strong topology-before-geometry claim. The place fields track the animal's current position rather than a future one: rebuilding each rate map on the position up to five steps ahead does not sharpen it at any prediction horizon, so the network's predictive character is expressed in its offline replay rather than in a spatial lead of its fields (though the discrete one-cell-per-step grid would not resolve a sub-cell anticipatory shift). All results use a single architecture (hidden size 500, prediction horizon $k = 5$) at one arena size; the horizon sweep shows the fold is present from $k = 1$ and the initialisation study varies the recurrent weights, but hidden size and arena size are not swept. The architecture has no excitatory–inhibitory separation. Finally, symmetry here is a property of the landmark layout, not of the arena geometry, which is square and therefore C_4-symmetric in every condition.

Outlook

The broader appeal of predictive learning is that a single, domain-general objective — predict the next observation — grows structured spatial representations with no hand-built spatial prior, in artificial networks as in the hippocampus (***Levenstein et al., 2024***; ***Stachenfeld et al., 2017***). What this study adds is that the geometry of the learned code is then set by the symmetry structure of the agent's interaction with the world — the group action of the environment on the self-motion signal — rather than by the architecture. That makes the paradigm a controlled developmental probe. Because the representation is learned rather than wired in, each factor that shapes it in an animal, and is there confounded or fixed, becomes a variable that can be turned on its own: the geometry of the enclosure, including irregular and non-convex ones; the information content and invariance of the self-motion signal, as here; the locomotor policy and the trajectory statistics it induces; and the task the agent is trained under. Charting how the code changes along these axes is a way to ask which features of an environment and an agent are responsible for which features of the representation.

Architecture is a variable of the same kind. The present network receives head direction as an oracle input and places no non-negativity constraint on its units, and it accordingly grows place-like but not grid-like tuning; hexagonal grids emerge in trained path integrators precisely when those constraints are imposed (***Sorscher et al., 2023***). Progressively more biological designs — a head-direction signal that is learned and allowed to drift rather than supplied, non-negative rates, separate path-integration and sensory streams, and modules for planning — would let one ask which ingredients are necessary for the joint emergence of place and grid cells and for the developmental order in which they appear. The aim of such a program is to turn the catalogue of spatial cell types into a set of consequences that follow from an objective, an architecture, and an experience, of which the symmetry-driven fold reported here is one.

Methods

Architecture and training.

We use `pRNN_th` (***Levenstein et al., 2024***) unchanged: a layer-normalised recurrent cell, hidden size 500, truncation $T = 200$, rollout horizon $k = 5$ unless stated, observation size 147 ($7 \times 7 \times 3$), action size 5. Training uses RMSprop ($\alpha = 0.95$, $\epsilon = 10^{-6}$) with four parameter groups at per-group learning rates $\lambda\sqrt{1/H}$ for $W_{\text{rec}}, W_{\text{out}}$, $\lambda\sqrt{1/(n_{\text{obs}} + n_{\text{act}})}$ for W_{in} and 0.1λ for the bias, with bias-only weight decay; $\lambda = 2 \times 10^{-3}$, batch size 8, gradient clipped per model to norm 1, 80,000 steps.

Arenas.

18×18 MiniGrid arenas (324 positions) with landmark layouts of C_1, C_2 and C_4 rotational symmetry. The observational discriminability index (ODI; the Spearman correlation between heading-averaged observation distance and spatial distance across all pairs of positions) is *not* matched across the three arenas, and cannot be: symmetric landmarks make distinct positions observationally identical, so discriminability necessarily falls with symmetry order. Measured over all 324 positions, ODI $= 0.336$ (C_1), 0.288 (C_2), 0.262 (C_4). An earlier report of these arenas described them as "matched at ODI $= 0.336$"; that value is the C_1 arena's ODI alone.

This does not confound the results, for two reasons. The decisive contrast — *axis* vs. *parity* — is made *within* one arena, on the same trajectories, so the observation stream and its ODI are identical between the two arms; the only difference is a 4×4 matrix on the heading block. And across arenas, ODI does not predict the readout: with a full compass, phase decoding is 0.981, 0.978 and 0.984 for C_1, C_2, C_4 (Pearson $r = -0.37$ against ODI, $p = 0.76$, $n = 3$ arenas), with the least discriminable arena scoring highest. Trajectories: 10,000 per arena, 200 steps, random start position and heading, random-walk policy.

Head-direction encodings.

The heading block of the action vector is left-multiplied by a fixed 4×4 matrix: identity (*full*); $\frac{1}{2}(I + P^2)$ where P is the cyclic shift (*axis*); the block-averaging matrix pairing $\{E, S\}$ and $\{W, N\}$ (*parity*); and $\frac{1}{4}\mathbf{1}\mathbf{1}^\top$ (*const*). Column sums are preserved, so activation magnitude is unchanged. Entropies under a uniform heading prior are $2, 1, 1, 0$ bits.

Ensemble training.

All networks sharing a computation graph (arena, encoding and seed all leave it unchanged; the horizon k does not) are trained as a single job using `torch.func.stack_module_state` with `vmap(functional_call)`, which lowers every per-model matrix multiply to one batched multiply and makes the kernel-launch count independent of the number of models. Forward passes are bit-exact against separate runs; gradients are bit-exact with noise disabled and agree to 2.3×10^{-10} absolute with the production noise on (`bmm` reassociation). Per-model gradient clipping, the four-group RMSprop and the sum-of-per-model-means loss are all preserved; each is checked against the reference implementation by test.

Orbit-phase decoding.

Hidden states are collected over 20,000 timesteps and averaged per position. Orbits are $\{x, R^2x\}$ under the $180°$ rotation (or the four-element C_4 orbits where stated); the arena side is even, so no rotation has an integer fixed point and every orbit has exactly $|G|$ cells. Phase labels are balanced, and cross-validation folds hold out whole orbits (`GroupKFold`), so the decoder must generalise a phase direction rather than memorise orbits. We report the accuracy of a logistic decoder, together with the R^2 of ridge decoders for the raw position and for the position within the fundamental domain. The latter is not a positive control — recovering a folded coordinate from an unfolded code requires a nonlinear map — so the sanity check that a code still represents space is $\max(R^2_{\text{raw}}, R^2_{\text{domain}})$.

Isotypic spectrum.

Each unit's rate map is decomposed into the four characters of C_4; P_k is the normalised power in character k, $\sum_k P_k = 1$, and $P_1 = P_3$ for real maps. RA $= P_0 - P_2$ and odd $= P_1 + P_3$.

Statistics.

All contrasts are two-sided exact Mann–Whitney U tests on per-network values. The planned contrasts are *axis* vs. *parity* within each arena, at $n = 10$ per cell for s1 and s2 and $n = 8$ for s4; at $n = 10$ vs. 10 the exact two-sided floor is $2/\binom{20}{10} = 1.1 \times 10^{-5}$, and a p-value at that floor indicates complete separation (every network in one group above every network in the other). Bonferroni correction across the three planned arena contrasts is applied where stated. Secondary experiments carry their own n: the prediction-horizon and noisy-compass sweeps use $n = 6$ per cell (floor 0.00216), the isotypic re-run $n = 5$ (floor 0.0079). Folded decoding accuracies are compared to their $1/|G|$ floor by a one-sample t-test where a residual is claimed. No results were excluded; models failing a criterion are reported as such rather than dropped.

Data and code availability

The source data underlying every figure and table are provided as comma-separated files in `project5_sym` a per-file index maps each reported value to its source (`Report/data/README.md`). All code, the analysis pipeline that regenerates those files from trained checkpoints, and the figure-generation scripts are openly available at https://github.com/RavulapalliManas/Thesis_work (`project5_symmetry/`), and will be archived on Software Heritage on acceptance. The test suite pins the claims that fail silently: the bit-exactness of the batched ensemble against per-model training, that the $k = 0$ target is exactly the anchor observation, that a synthetic C_2-folded code reads chance phase while a code with no spatial signal is distinguishable from it, and that the isotypic identity RA $= P_0 - P_2$ holds. Trained checkpoints and raw trajectories (~40 GB) are available from the author on request; every published number is reproducible from the committed data files without them.

Author contributions

M.V.S.R. designed the study, implemented the models and analyses, ran the experiments and wrote the manuscript.

Competing interests

The author declares no competing interests.

Use of AI tools

Large language models were used for code review, refactoring, and drafting assistance. All experiments, statistics and scientific claims were specified, executed and verified by the author; every number reported here is produced by the archived analysis scripts.

References

Burak Y, Fiete IR. Accurate path integration in continuous attractor network models of grid cells. PLoS Computational Biology. 2009; 5(2):e1000291. doi: 10.1371/journal.pcbi.1000291.

Calton JL, Stackman RW, Goodridge JP, Archey WB, Dudchenko PA, Taube JS. Hippocampal Place Cell Instability after Lesions of the Head Direction Cell Network. The Journal of Neuroscience. 2003; 23(30):9719–9731. doi: 10.1523/JNEUROSCI.23-30-09719.2003.

Gardner RJ, Hermansen E, Pachitariu M, Burak Y, Baas NA, Dunn BA, Moser MB, Moser EI. Toroidal topology of population activity in grid cells. Nature. 2022; 602(7895):123–128. doi: 10.1038/s41586-021-04268-7.

Grieves RM, Jenkins BW, Harland BC, Wood ER, Dudchenko PA. Place Field Repetition and Spatial Learning in a Multicompartment Environment. Hippocampus. 2016; 26(1):118–134. doi: 10.1002/hipo.22496.

Hartley T, Burgess N, Lever C, Cacucci F, O'Keefe J. Modeling place fields in terms of the cortical inputs to the hippocampus. Hippocampus. 2000; 10(4):369–379. doi: 10.1002/1098-1063(2000)10:4<369::AID-HIPO3>3.0.CO;2-0.

Knierim JJ, Kudrimoti HS, McNaughton BL. Place cells, head direction cells, and the learning of landmark stability. The Journal of Neuroscience. 1995; 15(3):1648–1659. doi: 10.1523/JNEUROSCI.15-03-01648.1995.

Langston RF, Ainge JA, Couey JJ, Canto CB, Bjaalie JG, Witter MP, Moser EI, Moser MB. Development of the Spatial Representation System in the Rat. Science. 2010; 328(5985):1576–1580. doi: 10.1126/science.1188210.

Leutgeb S, Leutgeb JK, Barnes CA, Moser EI, McNaughton BL, Moser MB. Independent codes for spatial context and spatial location in the hippocampal formation. Science. 2005; 309(5734):619–623. doi: 10.1126/science.1110711.

Levenstein D, Efremov A, Eyono RH, Peyrache A, Richards BA. Sequential predictive learning is a unifying theory for hippocampal representation and replay. bioRxiv. 2024; doi: 10.1101/2024.04.28.591528, posted 29 April 2024.

Momennejad I, Russek EM, Cheong JH, Botvinick MM, Daw ND, Gershman SJ. The successor representation in human reinforcement learning. Nature Human Behaviour. 2017; 1(9):680–692. doi: 10.1038/s41562-017-0180-8.

Muller RU, Kubie JL. The effects of changes in the environment on the spatial firing of hippocampal complex-spike cells. The Journal of Neuroscience. 1987; 7(7):1951–1968. doi: 10.1523/JNEUROSCI.07-07-01951.1987.

Sorscher B, Mel GC, Ocko SA, Giocomo LM, Ganguli S. A unified theory for the computational and mechanistic origins of grid cells. Neuron. 2023; 111(1):121–137.e13. doi: 10.1016/j.neuron.2022.10.003.

Spiers HJ, Hayman RMA, Jovalekic A, Marozzi E, Jeffery KJ. Place Field Repetition and Purely Local Remapping in a Multicompartment Environment. Cerebral Cortex. 2015; 25(1):10–25. doi: 10.1093/cercor/bht198.

Stachenfeld KL, Botvinick MM, Gershman SJ. The hippocampus as a predictive map. Nature Neuroscience. 2017; 20(11):1643–1653. doi: 10.1038/nn.4650.

Taube JS, Muller RU, Ranck JB. Head-direction cells recorded from the postsubiculum in freely moving rats. I. Description and quantitative analysis. The Journal of Neuroscience. 1990; 10(2):420–435. doi: 10.1523/JNEUROSCI.10-02-00420.1990.

Table 6. The model against the animal literature. "Reproduced" means the model shows the phenomenon without being fitted to it; "absent" means the phenomenon cannot exist in this architecture and we do not claim it.

observation in animals	source	in this model	verdict
Fields repeat across translationally related identical compartments; path integration does not separate them	***Spiers et al.*** (***2015***)	translation leaves the compass invariant, so the code folds onto X/G	reproduced
Repetition in parallel compartments, markedly less in radial ones; survives removal of the orientation landmark; attributed to self-motion	***Grieves et al.*** (***2016***)	rotation makes the compass equivariant, so the code lifts	reproduced
Multi-field firing increases with the rotational symmetry of the arena	***Muller and Kubie*** (***1987***); ***Knierim et al.*** (***1995***)	folding onto X/G makes one unit fire at all $\|G\|$ orbit-mates	reproduced
Head-direction input is required for a stable map; ADN lesions destabilise place fields	***Calton et al.*** (***2003***)	the *const* encoding folds the C_2 arena to near chance	consistent
Head direction plus recurrence plus prediction are needed for offline replay	***Levenstein et al.*** (***2024***)	offline trajectories exceed the coverage of a wake path of equal length only with the full compass and $k \geq 1$; coverage falls monotonically as the compass is ablated ($1.30 \rightarrow 0.80 \rightarrow 0.45 \rightarrow 0.31\times$ wake), and the autoencoder never leaves $0.37\times$. Replay appears at the same horizon as symmetry resolution and improves no further with a longer one	reproduced
Rate remapping: a slight rate-based discrimination between otherwise repeated fields	***Spiers et al.*** (***2015***)	absent. In the parallel arrangement the fields repeat perfectly (repetition $+0.997$) and the compartment is not decodable even by a nearest-neighbour readout at cells it has already seen (0.515, chance 0.5). The fold is complete; the animals retain a residual rate signal that this network does not	not reproduced
Grid cells on a toroidal population manifold, invariant across environments and sleep	***Gardner et al.*** (***2022***)	no grid cells (gridness ≤ 0.08); the torus cannot exist here	absent
Head-direction cells lie on a ring manifold that matures before place and grid coding	***Taube et al.*** (***1990***); ***Langston et al.*** (***2010***)	head direction is supplied as a four-point input, not learned	absent

Table 7. Where models of the spatial code stand on the symmetry dissociation. The decisive column is the last: the parallel and radial arrangements are locally identical, so any model that reads firing from local geometry or the current observation predicts the same repetition in both, and cannot produce the dissociation ***Grieves et al.*** (***2016***) observed.

model class	fields learned from experience	repetition in identical rooms	parallel-vs-radial dissociation
Continuous attractor / path integration (***Burak and Fiete, 2009***; ***Sorscher et al., 2023***)	metric built in	not addressed	no
Boundary-vector / geometry (***Hartley et al., 2000***)	partly	yes	no — predicts repetition in both
Successor representation (***Stachenfeld et al., 2017***)	yes (predictive)	depends on the given state space	not addressed
Reconstruction / autoencoder (this work, control)	yes	yes	no — folds both ($p = 0.59$)
Predictive RNN (***Levenstein et al., 2024***), this work	yes (emergent)	yes	**yes**, set by HD invariance